

AI 507: Artificial Intelligence and Society



#5 Innovation

Associate Prof. Dr. Lena Frischlich

Flashlight

Flashlight

Communication

complexity

episodic

contextual

inevitable

Lasswell formula

Who says what to whom in
which channel and with what effect?

Speech production

Stages

Conceptualisation

Formulation

Phonological encoding

Articulation

Intentional & goal
directed

Exhausting

Shortened and
referential

Text processing

Fast & quasi-automatic

Stroop test

Stages

Attention & Perception

Processing of the sign

Correct interpretation

Syntactical processing of
sentences

Inferential based on

Attribution of goals

Discourse schemes

Mental
Model

Limited capacity
model of motivated
mediated message
processing (LC4MP)

Communication in contexts

Communication accommodation
theory, CAT

Convergence vs. divergence

Overaccommodation vs.
underaccommodation

Computer-mediated
communication

Cues filtered out

Media richness theory

Cues filtered in

Social Information Processing
Theory (SIPT)
Hyper-personal model of
computer-mediated
communication
Social Identity
Deindividuation
Effects (SIDE)

AI-generated text: The transformer

AI: generated voice pro's & con's

Any questions to last week?



Then let's get started!

After today's lecture, I hope you will...

- ...have a better understanding of what an innovation is
- ...have discussed social innovations along the example of a current AI-election bot in development
- ...have explored the factors influencing whether an innovation will be accepted
- ...have discussed recent innovations – specifically around news

What is innovation?

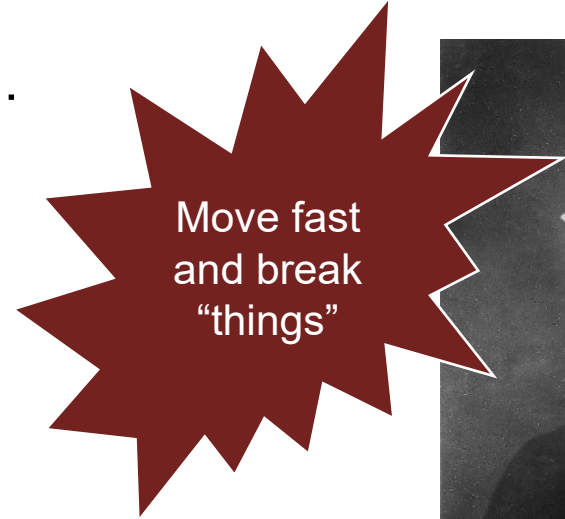
How can we define innovations?

(Carayannis, 2020; Hospers, 2005; Schumpeter, 1987)

Economists suggest that innovations are changes – these can be...

- technical
- social
- organizational
- scientific

Inventions lead to “creative destruction”
of the economic equilibrium



Joseph Schumpeter,
Unknown - Österreichische
Nationalbibliothek (Austrian National
Library), ID: 110D3539, Gemeinfrei,
<https://commons.wikimedia.org/w/index.php?curid=17120276>

Innovations are necessary because change is the normal state
of societies and social systems → regulation requires openness to innovation

Innovations can include

- New combinations
- New organizational structures
- New methods
- New contents or qualities
- New markets

Social innovations?

(Malhotra et al., 2021; <https://www.unesco.org/en/query-list/s/social-progress>)

- “[...] The process of developing and deploying effective solutions to challenging and often systemic social and environmental issues in support of social progress
- not the prerogative or privilege of any organizational form or legal structure.
- Solutions often require the active collaboration of constituents across government, business, and the nonprofit world.”

— Sarah A. Soule, Neil Malhotra, Bernadette Clavier, Center for Social innovation, Stanford

Social progress

= the advancement of society in terms of

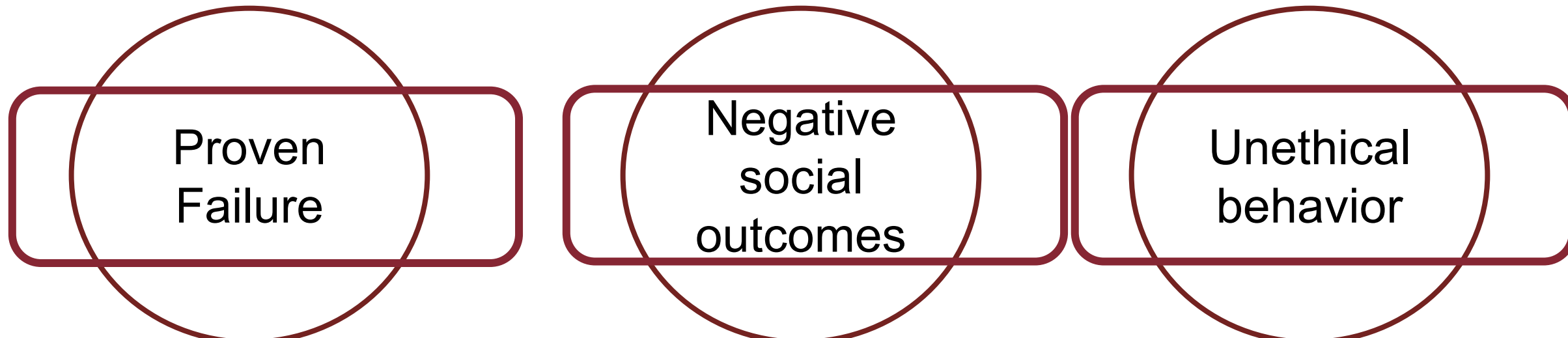
- social justice
- Equality
- overall well-being.

This concept encompasses improvements in various areas such as

- Education
- Healthcare
- human rights

=> collective commitment to enhance the quality of life for all

Three big NO's in (social) innovation (Malhotra et al., 2021)



Proven Failure

- some interventions have no impact at all
- Often because the actual customer/ user/ usage case has not been studied carefully

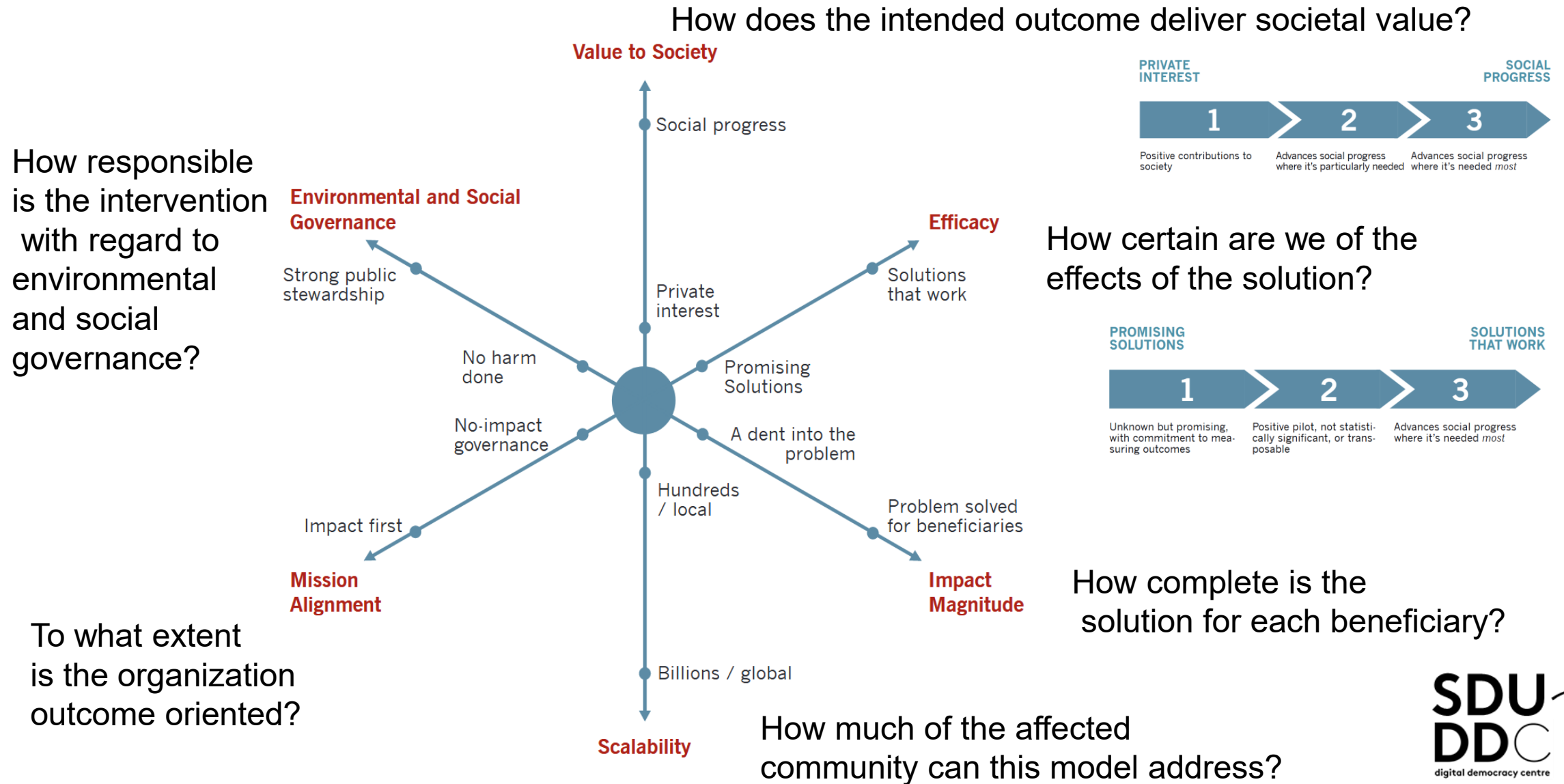
Negative social outcomes

- Well-meaning can still have severe negative consequences
- Either immediately (e.g., increased conflict/ pollution) or “downstream”

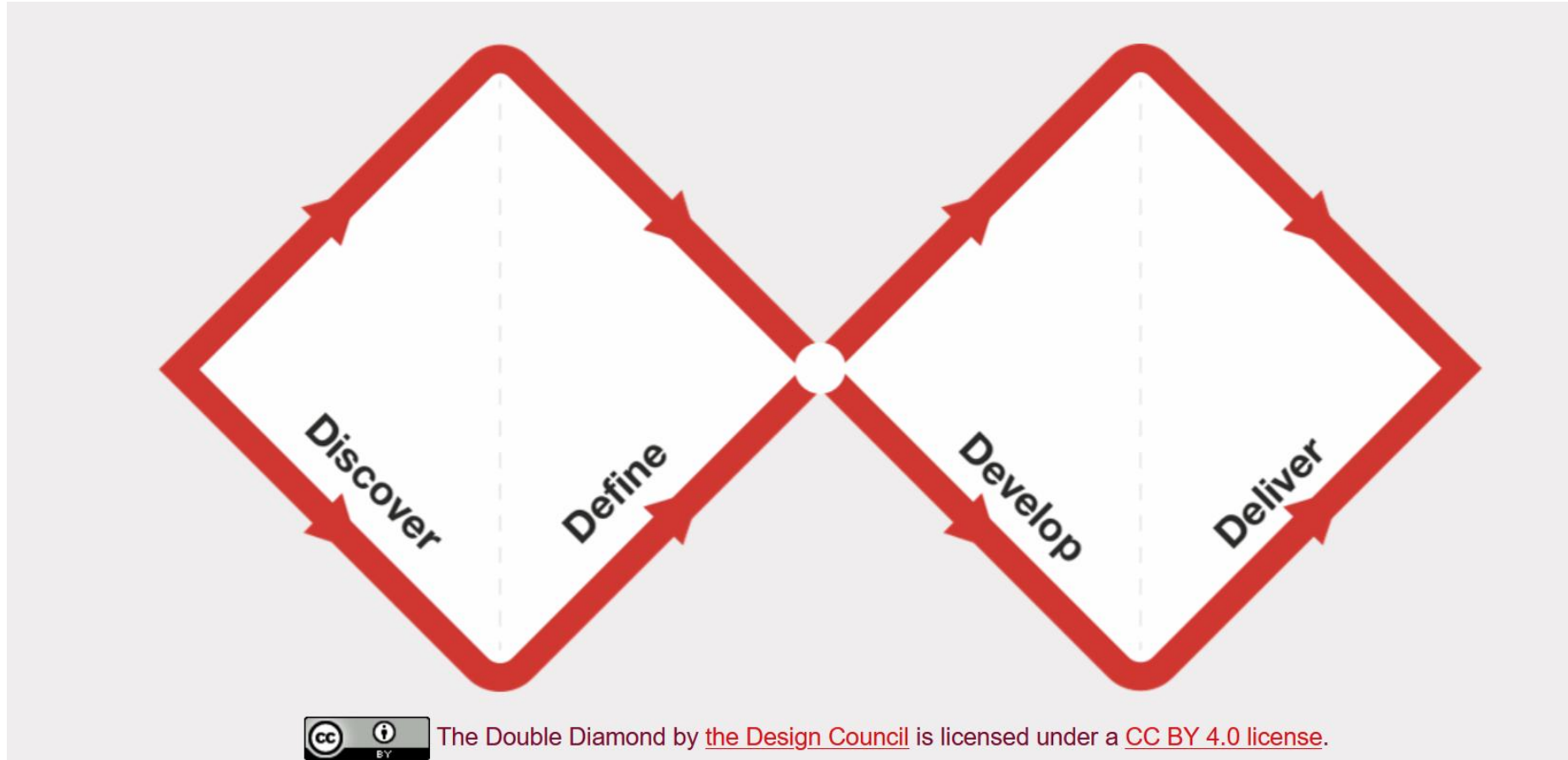
Unethical behavior

- Kant's golden rule
- Rawls “veil of ignorance”
- Utilitarianism
- Ethical principles of benevolence, non-maleficence & fairness

The social impact compass (Malhotra et al., 2021)



How do you get innovative?



Let's have a look at an innovation in the making



Emil Bakkensen Johansen

The AI Valgkompas FV26

1. Scan the QR code or go to app-ddc-fv26.cloud.sdu.dk

- App is in Danish, but you can force it to English if instructed: "Hi, please respond in English..."
- **NB: All your inputs are logged for debugging!**
- **DO NOT USE THIS AS VOTING ADVICE – it's a beta test**



2. Use the service for 5-7 minutes, talking about the Danish parties

AI Valgkompas FV26

Klar

AI:

14.01

Sådan bruger du AI-chatbotten

Chatbotten er designet til at give råd om hvilke tre partier, som stemmer bedst overens med dine holdninger. Kan du fortælle mig, hvilke emner du brænder mest for, eller hvilke værdier der er vigtigst for dig i politik? For eksempel: Er klima, økonomi, velfærd, immigration eller noget helt andet vigtigst?

Stil dit spørgsmål...

Afslut chat

Send

OBS: ALLE beskeder gemmes til forskning. Indtast ikke personlige oplysninger.

Vis disclaimer

The AI Valgkompas FV26

AI Valgkompas FV26

Klar

Alt: 14.01

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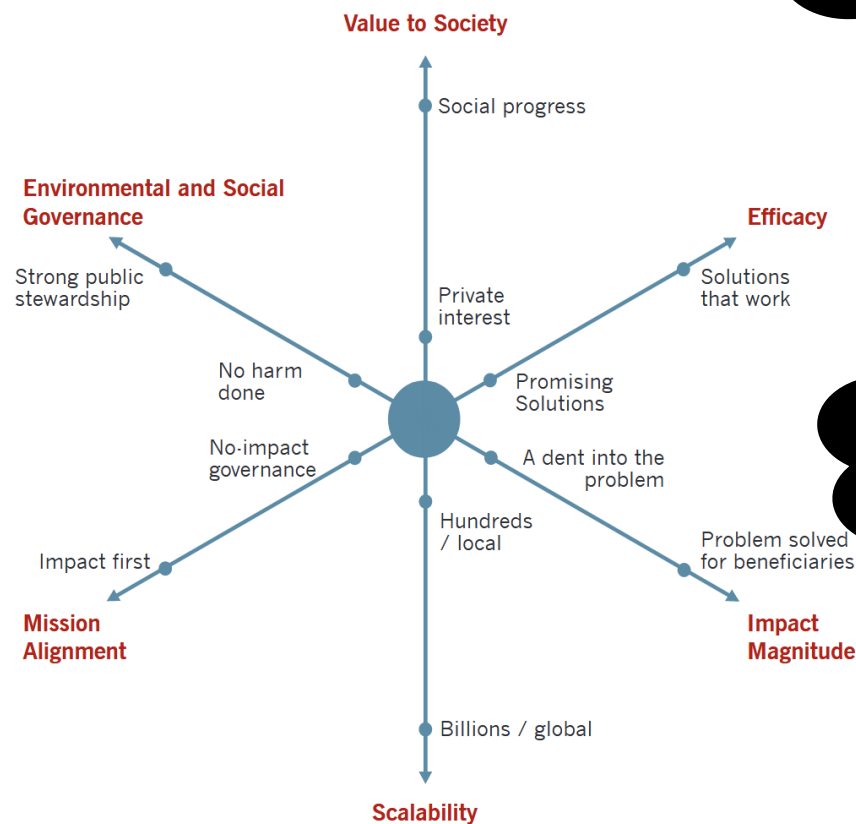
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Which kind of innovation is the Valgkompas?

How would you judge the Valgkompas using the social innovation compass?

Not all innovations are successful

<https://time.com/4704250/most-successful-technology-tech-failures-gadgets-flops-bombs-fails/>

failures-gadgets-flops-bombs-fails/

- The silicon valley spirit often calls for failing fast as failing is one of the best ways of learning
- Some innovations which have long landed on the graveyard of nostalgic memories yet have contributed substantially to what came after them
- Some of them were sold to competitors, some failed to keep innovating, some were caught by law

Will people use the valgkompass?
Why (not)?

20. Napster



Bruno Vincent—Getty Images

2. Virtual Boy



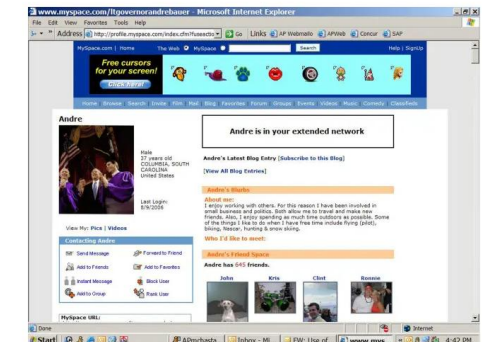
John Pfyke—Reuters

19. BlackBerry



Justin Sullivan—AP

12. MySpace



MARY ANN CHASTAIN—AP

What are your questions?



Then its time for a break!



**How do innovations
spread?**

Diffusion of innovations

(Everett cited from Rogers, 2005)

Innovation =

- A (new) idea or practice,
 - a (new) product,
 - a (new) technique or technology
- } that spreads through a social system
- Diffusion = process by which innovation is communicated through which channels over time between the members of a social system
 - To understand this diffusion, we have to analyze
 - how innovations are perceived [→ we must ask people]
 - where this perception comes from (e.g., mass media, culture) [→ e.g., content analyses]
 - how this perception influences the acceptance (or adoption) [→ e.g., experiments/surveys]

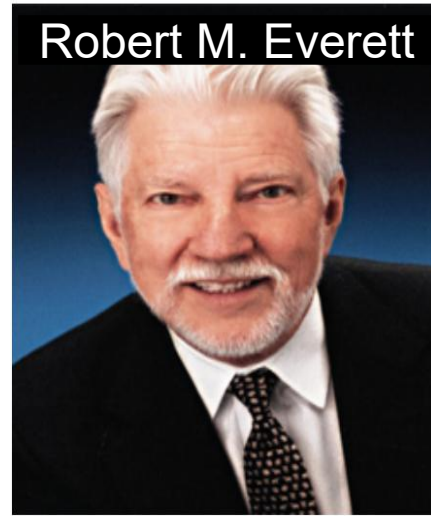


Photo Credit: Kim Jew Studio

Acceptance of innovations

(Ajzen & Fishbein, 1977; Ajzen, 2001; Davis, 1985)

- One broad and popular framework: Technology acceptance model by Fred Davis
- Rooted in the **theory of planned behaviour (TPB)** by **Ajzen & Fishbein**

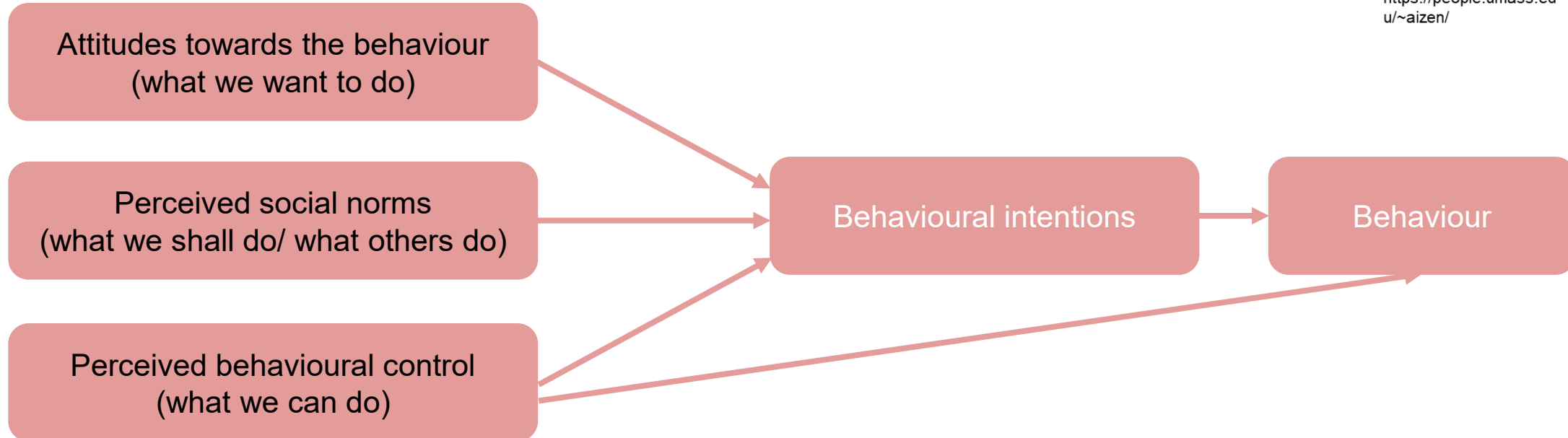


<https://people.umass.edu/~aizen/>



Martin Fishbein

<https://www.decisionsciences.com/?p=1198>



Acceptance of innovations

(Davis, 1985; Davis & Venkatesh, 1996; Feng et al., 2021; King & He, 2006)

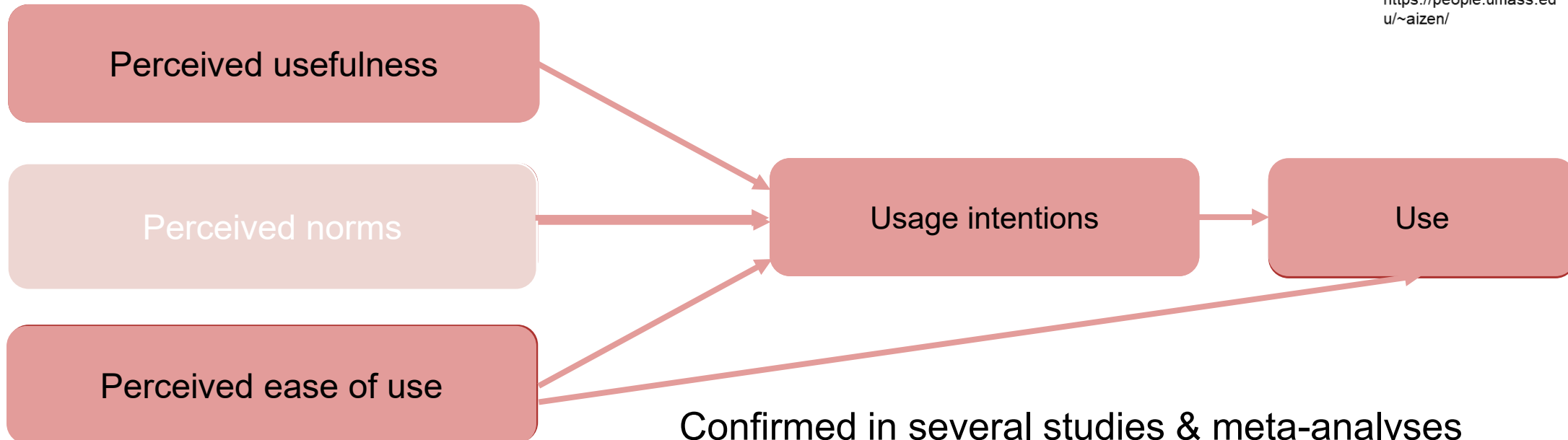
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<https://people.umass.edu/~aizen/>



<https://www.decisionsciences.com/?p=1198>



Confirmed in several studies & meta-analyses

Extension: Unified theory of acceptance and use of technologies

Acceptance as a result of ...

- Expectations regarding the performance of a technology (is it doing what it is supposed to?)
- Expectations regarding the (necessary) effort (what do I have to do)?
- Plus: social influence factors (Should I do that? Does everyone do it?)
- New: Further facilitating factors (e.g., trust)



The role of trust

(Bach et al., 2024)

- Trust = a risky action, a trustee is willing to risk an object, person, or process that he or she cannot control
- With full control, trust becomes unnecessary
- Central question: Whom and what can I trust?
- Remember the stereotype content model (→ #5 Social cognition): Two central dimensions of perception: Warmth/ Benevolence & Competence
- Competence, benevolence, and reputation → trustworthiness indicators
- Three main themes in the literature

Similarities to the dimensions described in the technology acceptance model/ theory of planned behaviour

Socio-ethical considerations

User characteristics

Technical and design features

Adaption is also a cognitive process (Vishwanath, 2015)

- Basic idea: Technology = cognitive construct that requires encoding, storage and retrieval
- New technologies can be more or less similar to other constructs in the mind (aka closer to existing “nodes”)
- If technologies are perceived to be similar to familiar or compatible with existing technologies, or related to positive concepts (my family & friends)
- More likely to be perceived as being useful
- Current studies often combine the different theories to provide a comprehensive cognitive, motivational, and communicative overview



What would these theories say about these innovations?



Chat GPT

- First language model based on the transformer architecture introduced in 2017
- 2019: Opening GPT-2 for the public
- 30.11.2022. GPT 3.5 gets a chat-based web-interface that is open for the public

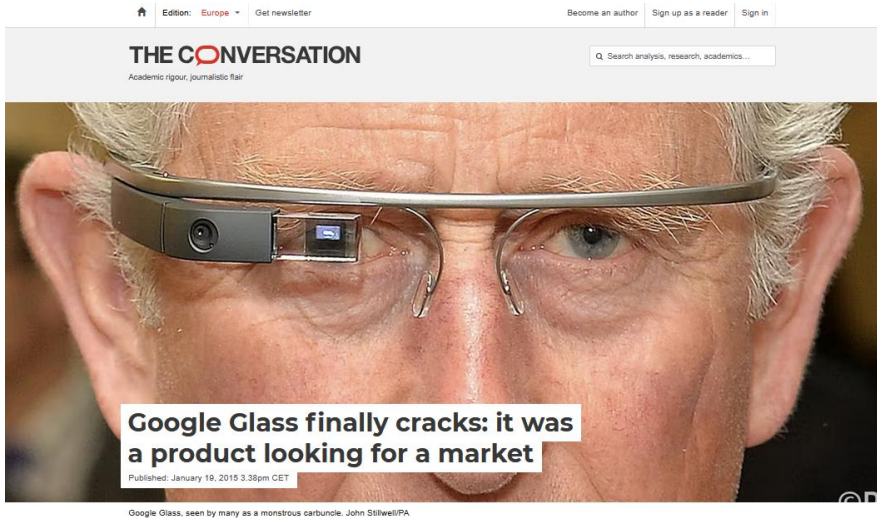


TikTok

- Precursor Douyin was founded in 2016
- Bought the popular platform music.ly in 2018 and created TikTok as successor
- The design was adapted to match the Chinese Douyin layout



What would these theories say about these innovations?



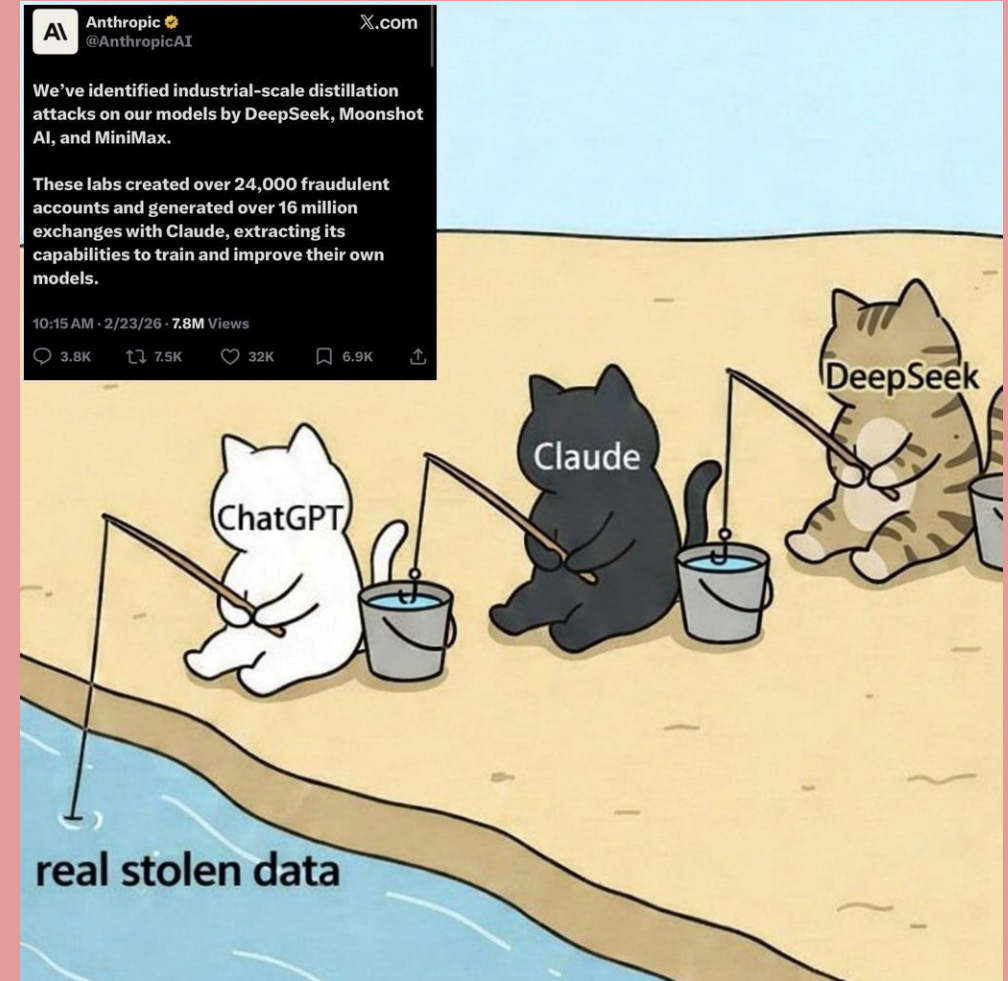
Google glasses

- Announcement 2012
- Available for developers April 2013 (1500\$)
- 2014 used at the Nanjing Olympic Games
- Privacy concerns, banned in several locations before release
- Safety concerns (e.g., during driving)



<https://www.meta.com/dk/ai-glasses/wayfarer/>

**Successful
technical/ business
innovations are
rarely “breaking
things” – they build
on existing work!**



https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://www.facebook.com/sarker.rj.7/photos/ha-this-meme-lands-pretty-well-the-irony-is-real-the-cats-fishing-from-the-same-/1600271164820450/&ved=2ahUKEwj17_Lt64WTaxVaVfEDHW7aHDYQh-wKegQIGRAE&usg=AOvVaw1AP3hm-PitE8rWbpH7bVYr

What are your questions?

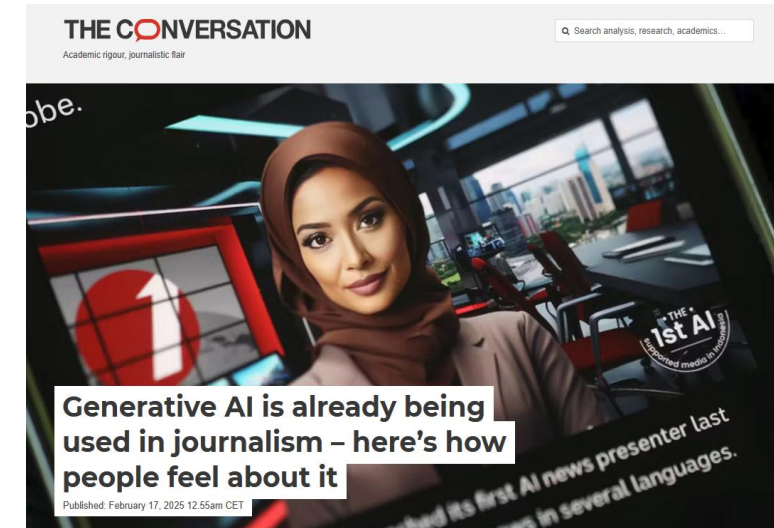


Example: Generative AI and journalism

News production covers – broadly – three phases:

(Thurman et al., 2019; Møller, 2023; Møller et al., 2025)

- Gathering
- Composing
- Distribution
- AI - and generative AI – has been trained on news extensively
- Also holds many promises for supporting journalists
- E.g., plays an increasing role in fine-tuning (e.g., headlines) and dissemination (e.g., through translation or adaption to different social media contexts)
- Still: Many challenges remain – due to biases, hallucinations, etc.
- Can lead to erosions of trust
- So: How do journalists use AI already today?

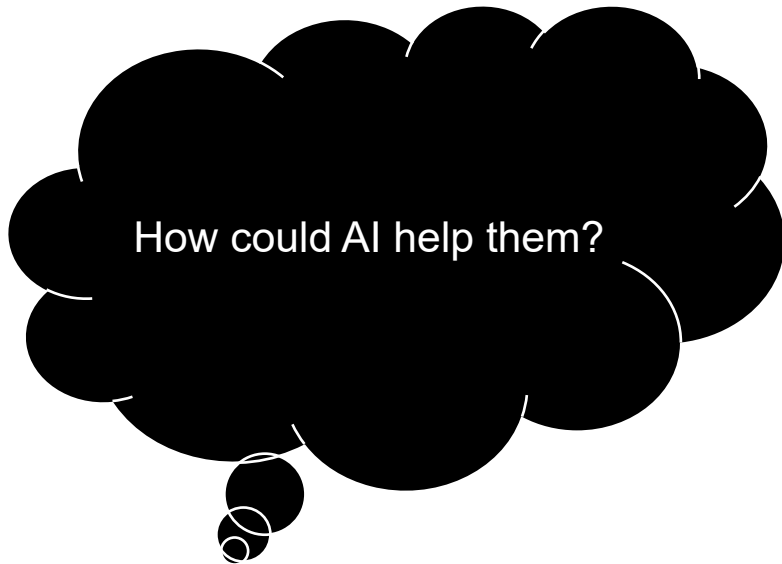


Indonesia's TVOne launched an AI news presenter in 2023. T.J. Thomson

Thomas et al., 2025

AI and news gathering (Stray, 2019)

- One central journalistic role is investigation
- Investigative journalists often work with *a lot* of data; they crawl through archives, call numbers, read tons of papers and drive to addresses to check if companies exist



- Rather seldomly used – mainly because training data is seldomly available, and the generalisability of models can be low
- Data is often not publicly available
- Accuracy must be extremely high to avoid liability
- But helpful for monitoring events, data extraction, pre-filtering and probably linking across databases

How (not to) run AI in investigative journalism

(de Lima Santos, 2022; Fridman et al., 2023; Young et al., 2018)

- Increasing relevance of data journalism and big-data analytics
- It has become more usual to rely on AI to create visualisations, verify information, analyse historical data, monitor social media and generate text
- Increasing relevance of multi-disciplinary teams → reliance on ready-available tools leads to struggles!
- Project in which news organisations could apply to get an AI professor from Oslo Met on their team for specific stories about: Petroleum reimbursement schemes/ drilling licenses, adverse events in elderly care, eating disorders in professional skiers (via computer vision), media landscape analysis in Norway (who publishes what where)
- Three central challenges:

1. Data Access

- Companies changing names
- Unstructured/missing data
- Privacy

2. Data Quality

- “real world coders”
- variation in data reporting between communities

3. Encoding newsworthiness

AI and fact-checking

(Dierickx et al., 2024)

Fact-checking includes different tasks

- Media monitoring
- Claim selection → what is checked
- Verification and evidence gathering
(! Not “falsification” but “confirmation”!)
- Giving a verdict
- Producing the story

LLM promises

- Media monitoring
 - Data extraction, processing and organisation
- Claim selection
 - Claim identification and ranking
- Verification and evidence gathering
 - Cross referencing claims, summaries of gathered information, data analysis
- Giving a verdict & producing the story
 - Summarizing the accuracy, providing context
generating text or supporting text production

BUT

Why fact-checking is not an AI task (Dierickx et al., 2024)

1. Hallucinations – and the lack of access to real-time data for retrieval-augmented generation
2. Source attribution errors
3. Bias in training data
4. Copyright issues and the threat of plagiarism
5. Wrongly summarized text
6. Risks of spreading misinformation that is overlooked due to the overconfidence in AI
7. Risk of reducing trust in fact-checking/ the media organisation
8. Risk of reducing human labour

Concerned about bullshitting machines
that learned how to sell bullshit
convincingly and a lack of laws against
bullshitting

(Sandra Wachter; Oxford Internet Institute, DGPUK 2025)

But: People increasingly use AI

(Newman et al., 2025; Waalen-Sandberg et

al., 2026)

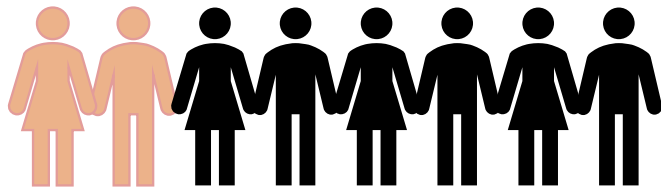
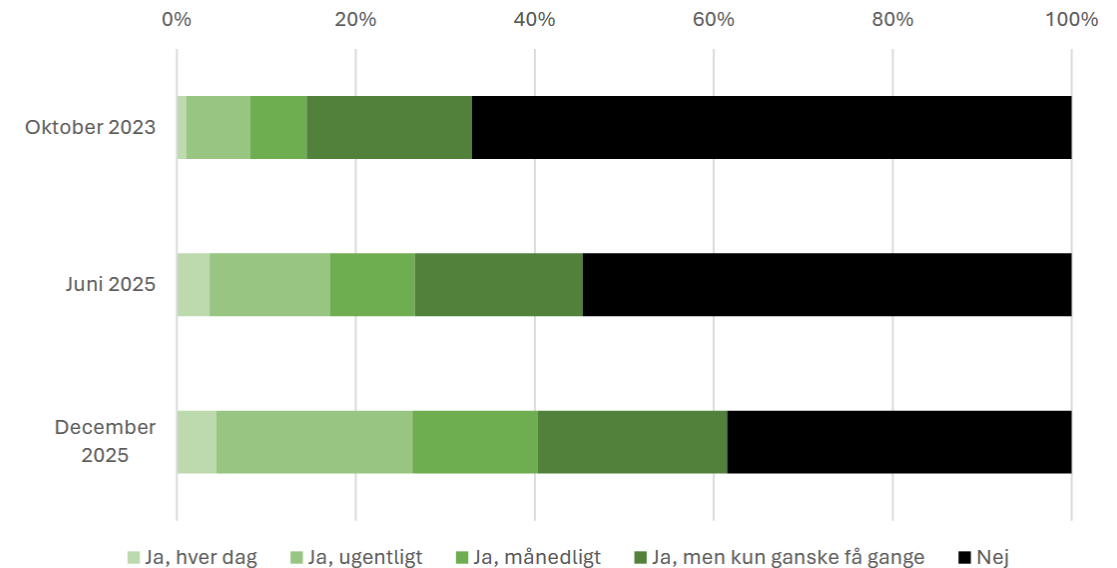
Digital Democracy Centre
Det Samfundsvidenskabelige Fakultet
Syddansk Universitet

sdu.dk/DDC

**Danskerne bruger mere AI, men er skeptiske over for
teknologiens udbredelse i samfundet**

Danskerne, digitalisering og kunstig intelligens 2022-2025

**Figur 10: Andel af danskere, der har benyttet generativ kunstig
intelligens i privat sammenhæng (2023-2025)**



24% (across countries) now use AI weekly
to get information

Still niche:

- Factual questions: 11%
- News: 6% (Japan & Argentina)
- More often: Making use easier to understand (~30%)

What are your questions?



Then I hope you now you...

- ...have a better understanding of what an innovation is
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- ...have explored the factors influencing whether an innovation will be accepted
- ...have discussed recent innovations – specifically around news

AI 507: Artificial Intelligence and Society



Have a great rest of the day!

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